MIL-C- 13891 (Ord)
23 December 1954
SUPERSEDING
PA-PD-610
13 October 1954

MILITARY SPECIFICATION
CASE, ROCKET, 2.75 INCH, T101E3

1. SCOPE

1.1 This specification covers one type of rocket case designated as Case, Rocket, 2.75 Inch, T101E3 for use with boosted rockets.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-M-151 - Metals, General Specification for the Inspection of.

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

P-85716 - Case, Rocket, 2.75 Inch T101E3. Assembly and Details.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 Rocket case.-The rocket case shall comply with all requirements specified in Drawings (dwg.) P-85716, and with all requirements specified in applicable specifications.

3.3 Pilot lot.-Prior to the initiation of production and whenever a change in materials or in manufacturing process occurs, a pilot lot representative of the process intended for use in production, shall be submitted. These cases shall be subjected to all the tests required for production to determine compliance with the requirements of this specification, applicable drawings. and specifications. No production lot shall be considered for acceptance until the pilot lot has been accepted. The manufacture of the production lot, prior to notification of acceptance of the pilot lot, shall be undertaken at no risk to the Government.

Submission of a pilot lot may be waived by the contracting officer where a new contract immediately follows the manufacture of satisfactory rocket cases of the same designation provided no change is made in materials or manufacturing process.

3.4 Workmanship.-All parts shall be free of chips, dirt, grease, rust and other foreign material. Parts which are assembled shall be thoroughly cleaned before assembly is performed. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of rocket cases produced by one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and shall be limited to the number of rocket cases which can be produced from one heat of steel. Prior to the start of production, a pilot lot of 100 rocket cases shall be made using the regular production line. The pilot lot shall be tested for compliance with this specification. The inspection lot for non-destructive inspection, within the above limitations, shall be designated by the inspector.

4.2 Sampling.

4.2.1 Pilot lot.

4.2.1.1 For mechanical properties tests.-Ten rocket case assemblies shall be selected from the pilot lot for test.

4.2.1.2 For hydrostatic test.-Ninety rocket case assemblies shall be selected from the pilot lot for test.

4.2.2 Production lot for nondestructive inspection.-Sampling for non-destructive inspection shall be conducted in accordance with Standard MIL-STD-105.

4.3 Inspection

4.3.1 General.-Inspection shall be as specified in Specification MIL-G-2550 and as specified herein.

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4.3.2 Packing and Marking.-The inspector shall ascertain that the packing and marking of the rocket case assemblied conform to this specification.

4.3.3 Classification of defects.-Inspection of the rocket case assemblies shall be as follows:

4.3.3.1 Case, Rocket, and Alternative, prior to plating (see dwg. P-85716).

Categories and defects	Gage or other method of inspection
Critical	
a. Hydrostatic test failure	Approved testing device
Major	
None	
Minor	
None	

4.3.3.2 Case, Rocket, and Alternative (see dwg. P-85716).

Categories and defects	Gage or other method of inspection
Critical	
a. Hydrostatic test stamp missing	Visual
Major	AQL 1.50 percent
1. Concentricity of front flange with tube	Gage
2. Concentricity of base flange with tube	Gage
3. Concentricity of igniter hole with counterbore	Gage
4. Concentricity of counterbore with tube	Gage
5. Concentricity of datum diameter at front flange angle with inside diameter of mouth	Gage
6. Parallelism of bottom of counterbore with base face	Gage
7. Perpendicularity of front flange face with tube	Gage
8. Perpendicularity of base with tube	Gage
9. Inside diameter of tube midsection	Gage
10. Inside diameter of forward section of tube for minimum (min) distance from mouth	Gage
11. Diameter of igniter counterbore	Gage
12. Total length	Gage
13. Depth of Igniter counterbore	Gage
14. Concentricity of outside diameter at 9.0 length with end 1.0 length outside diameter (dia)	Gage
15. Concentricity of inside diameter at 7.25 length with end 1.0 length outside dia.	Gage
16. Front flange not flush with tube (except P-85716A)	Straight edge
Minor	AQL 4.00 percent
41. Weight incorrect	Approved scale
42. Outside diameter of tube	Gage
43. Diameter of front flange	Gage
44. Diameter of base flange	Gage
45. Diameter of igniter hole	Gage
46. Wall thickness in midsection	Gage

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47. Thickness through base, min.----------Gage
48. Base metal exposed----------Visual
49. Boss inside flange missing----------Visual
50. Foreign matter----------Visual
51. Chamfer missing inside igniter hole----------Visual
52. Finish on front flange improper----------Visual
53. Finish improper----------Visual
54. Marking incorrect, incomplete or unidentifiable----------Visual

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Major
1. Hardness test failure----------Contractor's Certification
2. Tensile and elongation test failure----------Contractor's Certification
3. Salt spray test failure----------Approved testing device

4.4 Mechanical properties tests (pilot lot).

4.4.1 Procedure.

4.4.1.1 Hardness.-The representative rocket case assembly shall be tested for Rockwell hardness in accordance with Specification QQ-M-151. Observation shall be made to determine compliance with the requirements specified on the applicable drawing.

4.4.1.2 Tensile and elongation.-Using the same rocket case assemblies used in the hardness test, four tensile test specimens complying with Type 5A, Specification QQ-M-151 shall be taken along the longitudinal axis of each rocket case assembly. All specimens shall be tested in accordance with Specification QQ-M-151 to determine compliance with the requirements specified on the applicable drawing.

4.4.2 Rejection.-If any of the rocket case assemblies fail to comply with the mechanical properties requirements specified on the applicable drawing, the pilot lot shall be rejected without further test.

4.5 Hydrostatic test (pilot lot).

4.5.1 Procedure.-The representative rocket case assemblies shall be subjected to an internal hydrostatic pressure test. Each rocket case assembly shall be assembled in a fixture as shown on figure 1. The assembly shall then be subjected to the internal hydrostatic test specified on the applicable drawing.

4.5.2 Rejection.-If any of the rocket case assemblies fail to comply with the internal hydrostatic pressure requirement specified on the applicable drawing, the pilot lot shall be rejected without further test.

4.6 Mechanical properties tests (production lot).

4.6.1 Procedure.

4.6.1.1 Hardness.-A minimum of one rocket case assembly shall be selected from each lot and tested in the same manner as specified in 4.4.1.1 (see 6.1).

4.6.1.2 Tensile and elongation.-A minimum of one rocket case assembly shall be selected from each lot and tested in the same manner as specified in 4.4.1.2 (see 6.1)

4.7 Hydrostatic test (production lot).

4.7.1 Procedure.-Each rocket case assembly shall be tested as specified in 4.5.1.

4.7.2 Rejection.-If any failures occur, disposition of the lot shall be as specified in the appendix to Specification MIL-G-2550.

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5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking.-Packing, labeling and marking shall be as specified in Specification MIL-P-10025.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Certification shall be required to guarantee that hardness, tensile and elongation requirements specified on the drawing have been complied with (see 4.6).

Notice.-This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions is intended for use only in connection with procurement by the United States Government and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement nor used for any other purpose except when specifically authorized by the Chief of Ordnance.

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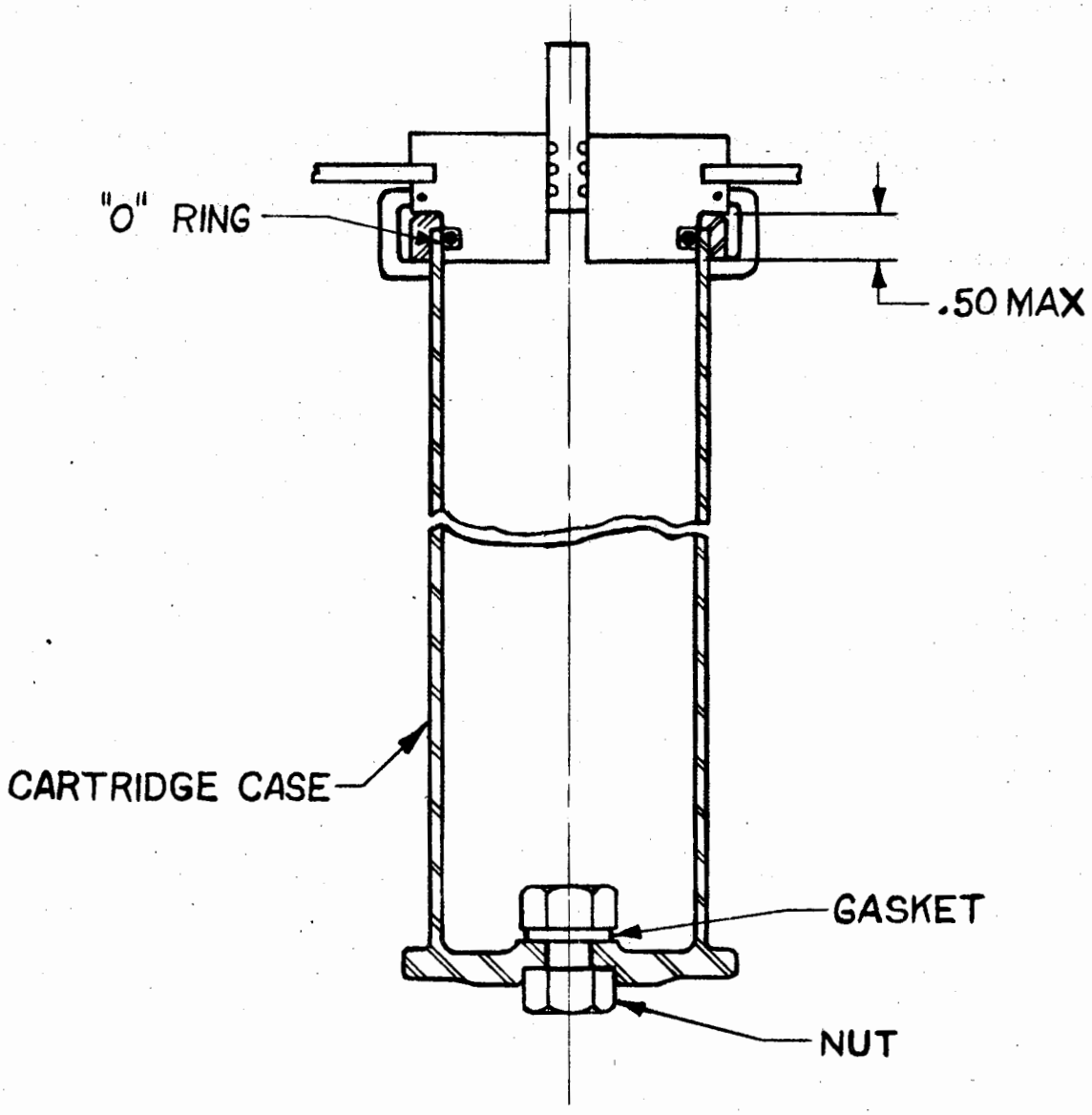


FIGURE 1

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